

Vehicle-Grid Integration Council Responses to CEC Load Shift Goal Questions

The Vehicle-Grid Integration Council will specifically focus its comments on utilizing electric vehicles (EVs) as load flexibility resources.

1. Are there any policy barriers in California impacting the full utilization of customer resources for load flexibility?

California faces several policy barriers reducing load flexibility potential from EVs:

Limited managed charging programs: Customers can best provide load flexibility services from an EV by having their charging sessions managed in different ways. This can be through price signals to encourage customers to engage in certain charging patterns or active managed charging programs where charging is coordinated and often controlled by communication signals sent from a utility, third-party aggregator, or other solution provider to a vehicle or charger.

Today, the majority of residential EV customers in California shape charging behavior according to utility time-of-use (TOU) rates. While simple and effective at reducing system peak demand, TOU rates don't maximize load-shifting potential and can cause "snapback effects" (sometimes called "timer peaks" or "secondary peaks"), where charging demand surges immediately after a TOU peak period. Though California investor-owned utilities (IOU) have recently launched new dynamic rate pilots in preparation for the CEC's Load Management Standards and designed under the CPUC's CalFUSE framework, real-time pricing rate design, platforms, and communication pathways are still under development. Although it is too early to determine whether these pilots will successfully attract EV customers, there are several barriers related to the underlying rate design, implementation, vendor selection, customer education, deployment of enabling technologies, and scalability that concern VGIC.¹

Need for diverse EV load flexibility and managed charging opportunities beyond real-time pricing pilots: Beyond TOU and dynamic rates, VGIC strongly recommends that California's EV owners and operators be offered other programmatic (i.e., non-rate) managed charging options, such as active managed programs focused on unlocking distribution system benefits, expanded emergency demand response programs such as the Demand Side Grid Support Program (DSGS) or the Emergency Load Reduction Program (ELRP), load modification contracts that create predictable EV load forecasts, and/or other new programs.² Diverse programs can also include different payment designs such as a fixed price per month for services provided, off-bill incentives, conditional treatment of charging costs (i.e., low- or no-cost off-peak charging in exchange for

¹ See <https://docs.cpuc.ca.gov/PublishedDocs/SupDoc/A2403019/7930/553184229.pdf>, page 15.

² Notably, legacy utility demand response programs like Base Interruptible Program and Capacity Bidding Program have prohibitive terms for EV participation (e.g., 6-hour dispatch events), and Demand Response Auction Mechanism is being discontinued.

dispatchable EV load flexibility), or other elements. Overall, it will be most beneficial for customers to be given a wide range of charging schedule and managed charging program options to find what best fits their specific needs and internal cost-benefit calculations. In its response to question 3 below, VGIC details the imperative to embrace a range of customer options to unlock meaningful load flexibility.

High interconnection costs for V2X resources: California has the highest DER interconnection application fees in the nation for small bidirectional charging systems. The IOUs' \$800 Rule 21 application fee for non-Net Energy Metering systems discourages the adoption of bidirectional chargers. This reduces the ability to use EVs for discharging to homes or buildings, a critical behind-the-meter load flexibility solution, in addition to net export to the grid to support system reliability and distribution system investment deferral and avoidance.

2. Do you recommend any successful market or regulatory models/approaches for utilizing load flexibility based on your experience outside California?

To date, California has largely focused on creating a load flexibility framework focused on reducing electric system peak demand. This is largely due to a focus on the system Resource Adequacy (RA) value of load flexibility. The focus on system-wide peaks is evident in the most popular load flexibility programs for EVs in California: TOU rates and emergency demand response programs such as DSGS or ELRP.

However, there are regulatory models and approaches that do not focus as much on utilizing load flexibility for system-wide peaks. Instead, there are increasing concerns about distribution congestion from EV charging causing “secondary peaks”, “timer peaks”, or “snapback effects” that fall outside of the system peak time.

To manage these secondary peaks, different jurisdictions are developing staggered charging approaches. These send tailored signals to individual EVs / EV supply equipment (EVSE) to distribute charging load more evenly across a given time period. For example, Baltimore General Electric (BGE) has partnered with WeaveGrid to deploy a managed charging program specifically focused on distribution asset protection. The program manages overnight charging load on individual circuits by sending signals to individual EVs on when to charge. BGE and Weavegrid assess total charging demand, individual battery state of charge, and individual charging times. Charging is then staggered across customers, prioritizing charging first for customers with the lowest state of charge and earliest departure time. The pilot has shown promising results, reducing peak demand on individual circuits by 33%.³

³ Smart Electric Power Alliance (SEPA), *The State of Managed Charging in 2024*, September 2024, pg. 43. Available at: <https://sepapower.org/resource/state-of-managed-charging-in-2024/>

3. Do you see any emerging trends in California’s market and electricity consumers that may impact the state’s load flexibility potential?

Every year, California adds more EVs to its roads, and the amount of EVs is expected to greatly increase in the next decade as the Advanced Clean Cars, Advanced Clean Trucks, and Advanced Clean Fleets regulations from the California Air Resources Board (CARB) encourage and require additional EV sales. CARB estimates that in order to meet AB 32 goals of carbon neutrality in 2045, California will need 105,555 GWh of electricity annually to power over 22 million light duty and 1.2 million medium and heavy-duty vehicles.⁴ Notably, this modeling was conducted in 2022, before the adoption of Advanced Clean Fleets. The sheer amount of EVs on the road, in addition to other newly electrified load, will greatly increase electricity demand but will also create additional load flexibility potential on the grid, especially if customers are encouraged to participate in VGI programs.

While there is ample evidence that EV charging preferences are especially diverse, VGIC expects that preferences will also change over time. For example, the EV market is currently dominated by light-duty consumers that have access to private charging at their home, and most charging is currently done at single-family residences.⁵ However, many Californians, particularly many renters, lack access to home charging and will depend on public chargers. Business and municipal fleets across light-, medium-, and heavy-duty vehicle classes are also currently exploring how to best schedule and manage charging for their own needs. These mobility needs are very different from private passenger vehicles and extremely diverse across sectors – even more diverse than existing commercial and industrial load. For example, Southern California Edison has shared the following range of EVSE power levels by sector, which ranges from 1.5 kW to 1,200 kW.⁶ Notably, this data is collected from the nation’s largest medium- and heavy-duty make-ready program.

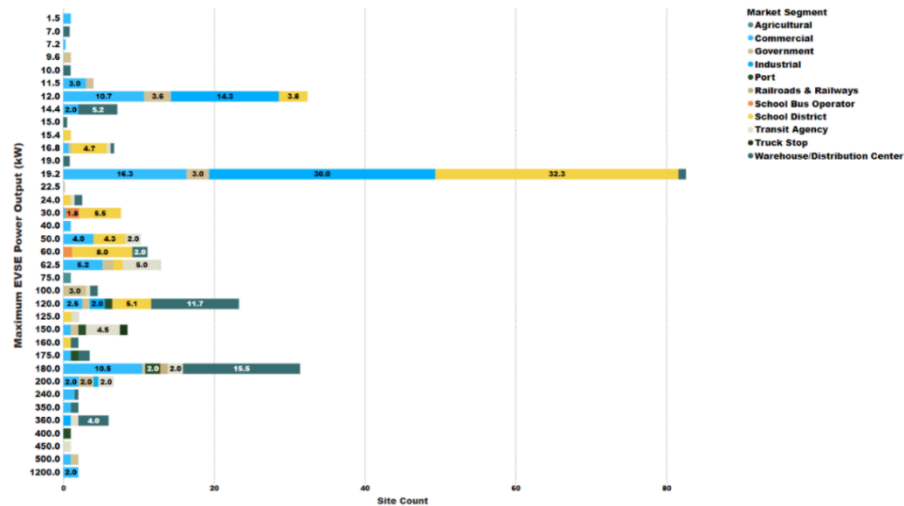
⁴ CARB, 2022 Scoping Plan. Available at: <https://ww2.arb.ca.gov/sites/default/files/2023-04/2022-sp.pdf>

Data available here: <https://ww2.arb.ca.gov/sites/default/files/2022-11/2022-sp-PATHWAYS-data-E3.xlsx>

⁵ Plug In America, 2024 EV Driver Annual Survey Report, pg. 15. Available at: https://pluginamerica.org/wp-content/uploads/2024/09/2024-Plug-In-America-EPRI-EV-Driver-Survey-Report_Final.pdf

⁶ Southern California Edison, Transportation Electrification Program Advisory Council, December 13th, 2024, pg. 14 (Charge Ready Transport).

EVSE Power Output Levels by Market Segment



Note, figures reflect active, committed sites with signed CRT Program Agreements. Partial site counts represent sites with multiple types of charging equipment.

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Given the diversity in EV charging behavior across both residential and non-residential customers, a wide range of managed charging opportunities will be needed to meet different customer needs. VGIC reiterates its recommendation that customers be provided with a menu of EV load flexibility opportunities from which to choose based on their vehicle needs, preferences, and internal cost-benefit calculations.

4. Do you have any other conceptual ideas or feedback for the state to consider?

To unlock the full potential of EV demand flexibility, ensure an effective focus on vehicle-grid integration (VGI), and track policy success and adjust accordingly, the CEC should establish a distinct EV load flexibility or VGI goal. This goal would recognize that EVs, as a dual-purpose resource providing both transportation and grid services, require tailored policies and programmatic approaches that differ from other load flexibility technologies like smart thermostats or electric appliances. A dedicated goal for VGI would allow for a nuanced approach that accounts for these diverse needs, ensures that mobility needs are met, and considers the broader greenhouse gas and air pollution reductions that come from transportation electrification. This goal should be defined in megawatt (MW) contributions of EVs to peak load reduction, and further segmented into MW contribution from unidirectional charging equipment and MW contribution from bidirectional charging equipment. The target could also include or incorporate metrics such as a target percentage of EVs participating in managed charging and/or bidirectional charging programs (not including TOU rates, which is assumed to be a baseline behavior per the CEC's Integrated Energy Policy Report) or a specified level of bidirectional charging equipment adoption by a given date. VGIC offers itself as a resource to support the development of such a target, either within the load shift goal or as part of another CEC proceeding or initiative.

To support achieving this goal, the CEC should require IOUs and large publicly-owned utilities to develop programs to unlock widespread EV load flexibility. Just as the CEC's LMS directs these utilities to offer optional dynamic rates to all customer in 2027, the CEC's new VGI target should direct these utilities to offer optional managed charging and/or bidirectional charging program options to all customers in 2027, or sooner. Given California's ratepayer affordability crisis and restricted state budget, VGIC recommends the CEC direct these utilities to initially leverage non-ratepayer funding, which could include Greenhouse Gas Reduction Funds and/or Clean Transportation Program funds. Over time, as program cost-effectiveness and success against the established VGI target are assessed, the utilities may transition to using ratepayer funding in a manner that ensures offerings yield net ratepayer benefit and provide a sustainable, reliable regulatory signal to customers and VGI solution providers.